Dissertation Title:
Predictive Analytics Framework for Demand Forecasting

Course No.: SEZG628T

Course Title: Dissertation

Dissertation Done by:

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Degree Program: M. Tech

Research Area: ARTIFICIAL INTELLIGENCE

Dissertation carried out at:

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1. Broad Area of Work

Predictive analytics and demand forecasting are important research areas in the current era of data-driven decision-making. Organizations increasingly rely on historical data and machine learning techniques to predict future demand and optimize operational planning. The broad scope of this project includes concepts from data analytics, statistical modeling, and machine learning.

The project focuses on the following application areas:

- **Time-series analysis:** To capture trends, seasonality, and cyclic patterns in historical data
- **Predictive analytics:** To forecast future demand based on past observations
- **Machine learning:** To learn complex patterns and improve prediction accuracy
- **Decision support systems:** To assist operational planning and resource optimization

2. Background

The adoption of flexible work arrangements has introduced new operational challenges for organizations, particularly in managing on-site infrastructure and employee facilities. In environments where employees attend the office for a limited number of days per week and workstations are not pre-assigned, accurately estimating daily office occupancy becomes critical for effective resource planning. One such challenge is predicting the number of employees utilizing shared services such as office canteens, where inaccurate forecasts can lead to food wastage or service delays.

This project proposes an end-to-end predictive analytics framework to forecast daily employee attendance using historical breakfast consumption data as a proxy for office presence. Since breakfast participation consistently represents a high proportion of on-site employees, it provides a practical and privacy-preserving alternative to sensitive access control or swipe-card data. The model incorporates temporal patterns and contextual factors such as weekdays, long weekends, seasonal trends, and end-of-year leave periods to improve forecast accuracy.

The primary objective of the proposed solution is to assist canteen operations in preparing appropriate quantities of food, thereby minimizing wastage and ensuring timely service. The framework also demonstrates potential value for broader facility planning, including desk allocation and logistical preparedness. Although motivated by a real-world corporate environment operating under a flexible work model, the proposed

approach is generic and can be extended to other organizations with similar hybrid or flex-work policies. The outcomes of this project aim to support data-driven decision-making while promoting operational efficiency and sustainability.

3. Objectives

The objectives of this project are as follows:

- To study attendance patterns in flexible and hybrid work environments
- To analyse historical breakfast attendance data as a proxy for employee inflow
- To identify temporal and calendar-based factors influencing daily attendance
- To design and implement predictive models for next-day attendance forecasting
- To assist canteen operations in reducing food wastage through accurate demand estimation
- To propose a generalized demand forecasting framework applicable to similar organizational contexts

4. Scope of Work

The scope of this dissertation includes:

- Analysis of historical breakfast attendance data
- Data preprocessing and exploratory analysis to identify trends and patterns
- Feature engineering to capture effects of weekdays, holidays, long weekends, and seasonality
- Development and evaluation of statistical and machine learning-based forecasting models
- Performance comparison using appropriate forecasting error metrics

The scope of this project is restricted to offline data analysis and next-day prediction using historical breakfast consumption records. The solution does not utilize sensitive employee access logs, attendance systems, or personally identifiable data, ensuring privacy and compliance with organizational data policies.

While the proposed framework can be integrated with canteen planning workflows to support food preparation and inventory decisions, real-time system integration and the

use of official office attendance or access-control data are outside the scope of the current work. Based on the initial results and effectiveness of the model, the solution may be proposed for broader organizational adoption in the future; however, such extensions are not considered within this project's defined boundaries.

5. Plan of Work

Phases	**Start Date-End Date**	**Work to be done**
Dissertation Outline	15 Jan 2026 – 05 Feb 2026	Literature Review and prepare Dissertation Outline (Study of demand forecasting and predictive analytics techniques)
Data Understanding	06 Feb 2026 – 15 Feb 2026	Data exploration and preprocessing
Feature Engineering	15 Feb 2026 – 28 Feb 2026	Modeling calendar and seasonal effects
Model Development	01 March 2026 – 19 March 2026	Implementation of forecasting models
Model Evaluation	20 March 2026 - 09 April 2026	Validation and performance analysis
Dissertation Review	10 April 2026 - 30 April 2026	Dissertation writing and final review (Submit Dissertation to Supervisor & Additional Examiner for review and feedback)
Submission	1 May 2026 – 10 May 2026	Final Review and submission of Dissertation

6. Literature References

The literature review focuses on demand forecasting, time-series analysis, and predictive analytics. The following references are considered:

1. *G. Box, G. Jenkins, and G. Reinsel, Time Series Analysis: Forecasting and Control, Wiley, 2015.*
2. *R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, OTexts, 2018.*
3. *Makridakis, S., Spiliotis, E., & Assimakopoulos, V., "Statistical and Machine Learning Forecasting Methods," International Journal of Forecasting, 2018.*

7. Particulars of the Supervisor and Examiner

	Supervisor	**Additional Examiner**
Name	Jaydeep Gaikwad	Sandeep Jain
Qualification	M. Tech	B. Tech
Designation	Senior Manager R&D Verification	Sr. Staff Engineer R&D
Employing Organization and Location	Ansys (Part of Synopsys), Pune	Ansys (Part of Synopsys), Pune
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8. Remarks of the Supervisor

The dissertation outline presents a clear and focused problem statement in the domain of predictive analytics with direct applicability to operational planning in a hybrid work environment. The objectives, scope, and plan are well-defined. The consideration of temporal, calendar-based, and seasonal factors in the forecasting approach demonstrates sound analytical thinking. The proposed work plan is realistic and achievable within the given timeframe. I find the outline satisfactory and recommend it for approval.

Information about the Supervisor:

Name: Jaydeep Gaikwad

Education:

- B.E. (Mechanical) - Walchand College of Engg. Sangli 1996
- M.Tech (Solid Mechanics and Design) – IIT Kanpur (1999-2001)

Work Experience:

- Bajaj Auto Ltd. Aurangabad – 1996-97 (Graduate Trainee Engineer)
- Sanjivani College of Engineering Kopargaon – 1997-1999 (Lecturer)
- Presently working as Senior Manager R&D Verification at Ansys Inc (joined in 2001)

Publications: Static equilibrium analysis of compliant mechanical systems using relative coordinates and loop closure equations

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2021-2022

SEZG628T: Dissertation

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DISSERTATION / PROJECT / PROJECT WORK TITLE	Predictive Analytics Framework for Demand Forecasting

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Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing organization)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
Name: Patil Kiran Sanjay	**Name: Jaydeep Gaikwad**	**Name: Sandeep Jain**